

ME698V: Physics-Based Modeling of Li-ion Batteries

Mid-Term Project

Development of a Single-Particle Model (SPM) for
Carbon | LiPF₆ Electrolyte | NMC111 Li-ion Cells

Project Statement

A commercial electric vehicle (EV) manufacturer requires a mathematical model for the Carbon | LiPF₆ Electrolyte | NMC111 cell used in their EV powertrain. A single-particle model (SPM) is developed in three stages: (i) the Carbon | LiPF₆ | Li half-cell is modeled following Verbrugge and Koch's thermodynamic interaction formalism; (ii) the NMC111 | LiPF₆ | Li half-cell is modeled with an analogous formalism and its interaction parameters fit to the reported voltage-capacity data; (iii) the two half-cell models are combined into a full-cell response, first at equilibrium and then including Butler–Volmer kinetics at the particle surface.

1 Carbon Anode Half-Cell

1.1 Chemical potential and governing equation

Following Verbrugge and Koch, the chemical potential of intercalated lithium in the carbon host is

$$\mu_{\text{Li}} = \mu_{\text{Li}}^0 + RT \ln \left(\frac{c}{1-c} \right) + \sum_{k=2}^7 k \Omega_k^a c^{k-1}, \quad 0 \leq c \leq 1, \quad (1)$$

where c is the fractional occupancy and Ω_k^a are the anode interaction coefficients. The flux is driven by the chemical-potential gradient,

$$N = -\frac{Dc}{RT} \nabla \mu_{\text{Li}} = -D \left(c \frac{d\mu_{\text{Li}}}{dc} \right) \nabla c. \quad (2)$$

Differentiating,

$$c \frac{d\mu_{\text{Li}}}{dc} = \frac{RT}{1-c} + \sum_{k=2}^7 k(k-1) \Omega_k^a c^{k-1}, \quad (3)$$

so that $N = -D_{\text{eff}}(c) \nabla c$ with the thermodynamic-factor-corrected diffusivity

$$D_{\text{eff}}(c) = D \left[\frac{1}{1-c} + \frac{1}{RT} \sum_{k=2}^7 k(k-1) \Omega_k^a c^{k-1} \right]. \quad (4)$$

For the single particle (spherical geometry, radial transport only), the material balance is

$$\frac{\partial c}{\partial t} = -\nabla \cdot N = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 D_{\text{eff}}(c) \frac{\partial c}{\partial r} \right). \quad (5)$$

1.2 Boundary and initial conditions

Symmetry at the particle center and continuity of the intercalation flux with the imposed current density i_{app} at the surface ($i_{\text{app}} > 0$ denotes Li insertion / lithiation) give

$$\left. \frac{\partial c}{\partial r} \right|_{r=0} = 0, \quad (6)$$

$$\left. \frac{\partial c}{\partial r} \right|_{r=R} = \frac{i_{\text{app}}}{F D_{\text{eff}}(c_s)}, \quad c_s \equiv c(R, t), \quad (7)$$

with initial condition $c(r, 0) = c_0 = 5.0 \times 10^{-4}$ (uniform).

1.3 Cell voltage

With the exchange current density taken large compared to i_{app} ($\eta \approx 0$), the cell voltage relative to a Li metal reference is

$$V = U^s + \frac{RT}{F} \ln \left(\frac{1 - c_s}{c_s} \right) - \frac{1}{F} \sum_{k=2}^7 k \Omega_k^a c_s^{k-1}. \quad (8)$$

1.4 Parameters and implementation

Table 1 lists the parameters used, taken from Tables I and II of Verbrugge and Koch (1996). Equations (4)–(8) were implemented in PyBaMM as a custom `BaseModel` on a spherical particle domain, discretized with the finite-volume method (50 nodes) and integrated with the default solver for $i_{\text{app}} = +1.0$ mA/cm².

Table 1: Carbon anode parameters (Verbrugge and Koch, 1996).

Parameter	Symbol	Value
Particle radius	R	3.5 μm
Site concentration	c_{max}	18 000 mol/m ³
Guest diffusivity	D	1×10^{-14} m ² /s
Standard potential	U^s	0.8170 V
$\Omega_2^a/F, \dots, \Omega_7^a/F$	—	0.9926, 0.8981, −5.630, 8.595, −5.784, 1.468 V

The simulated charging curve reproduces the experimental data of Fig. 1(a) closely across the full occupancy range (Fig. 1), confirming that the thermodynamic-factor-corrected diffusivity of Eq. (4) combined with the equilibrium voltage relation of Eq. (8) captures the strongly non-ideal intercalation behavior reported by Verbrugge and Koch.

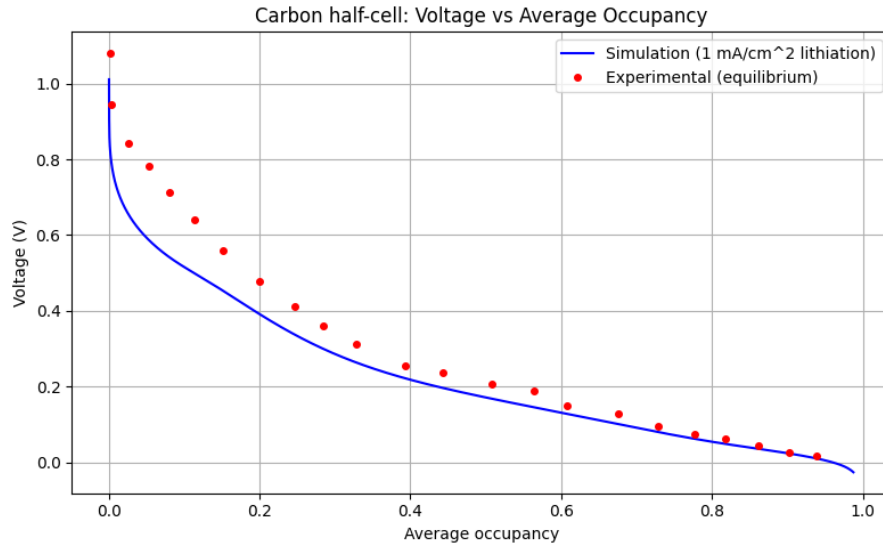


Figure 1: Simulated and experimental (Table 1) voltage vs. average occupancy for the carbon half-cell, $i_{\text{app}} = 1.0 \text{ mA/cm}^2$.

2 NMC111 Cathode Half-Cell

2.1 Chemical potential and governing equation

The chemical potential of lithium in NMC111 is written with the interaction sum evaluated in the vacancy fraction,

$$\mu_{\text{Li}} = \mu_{\text{Li}}^0 + RT \ln \left(\frac{c}{1-c} \right) - \sum_{k=2}^7 k \Omega_k^c (1-c)^{k-1}, \quad (9)$$

where c is again the fractional occupancy of Li in the host. Proceeding exactly as in Section 1,

$$c \frac{d\mu_{\text{Li}}}{dc} = \frac{RT}{1-c} + \sum_{k=2}^7 k(k-1) \Omega_k^c c (1-c)^{k-2}, \quad (10)$$

so the effective diffusivity for the cathode particle is

$$D_{\text{eff}}(c) = D \left[\frac{1}{1-c} + \frac{1}{RT} \sum_{k=2}^7 k(k-1) \Omega_k^c c (1-c)^{k-2} \right]. \quad (11)$$

Note the extra factor of c multiplying $(1-c)^{k-2}$: because the interaction sum in μ_{Li} is a function of $(1-c)$ rather than c , the chain rule introduces this asymmetry relative to the anode's Eq. (4), and the two expressions are *not* related by the naive substitution $c \rightarrow (1-c)$. The governing PDE and symmetry condition at $r = 0$ retain the same form as Eqs. (5) and the first line of (7). The surface flux condition, initial condition, and voltage expression are

$$\left. \frac{\partial c}{\partial r} \right|_{r=R} = \frac{i_{\text{app}}}{F D_{\text{eff}}(c_s)}, \quad c(r, 0) = 0.99, \quad (12)$$

$$V = V_0 + \frac{RT}{F} \ln \left(\frac{1-c_s}{c_s} \right) + \frac{1}{F} \sum_{k=2}^7 k \Omega_k^c (1-c_s)^{k-1}. \quad (13)$$

With $i_{\text{app}} = -1.0 \text{ mA/cm}^2$ (Li^+ leaving the cathode), c_s decreases from 0.99 and the specific capacity referenced to the NMC111 mass is

$$Q_{\text{sp}}(t) = \frac{F c_{\text{max}}}{3600 \rho_{\text{NMC}}} (c_0 - c_s(t)) \text{ [mAh/g]}, \quad \frac{F c_{\text{max}}}{3600 \rho_{\text{NMC}}} = 609.7 \text{ mAh/g} \equiv Q_{\text{max}}. \quad (14)$$

2.2 Parameter fitting

V_0 and Ω_k^c ($k = 2, \dots, 7$) were obtained by a constrained nonlinear least-squares fit (`scipy.optimize.curve_fit`) of the equilibrium ($\eta = 0$) voltage–capacity relation, Eqs. (13) and (14), to the twenty data points of Table 2, with bounds fixed to the ranges quoted in the project statement:

$$V_0 \in [3.0, 4.0], \quad \Omega_2^c/F \in [-0.15, 0], \quad \Omega_3^c/F \in [1.5, 2.0], \quad \Omega_4^c/F \in [-3.2, -2.5], \\ \Omega_5^c/F \in [10.0, 14.0], \quad \Omega_6^c/F \in [11, 15], \quad \Omega_7^c/F \in [-4.0, -2.5].$$

The fit converged to the values in Table 2, with an RMS residual of 22 mV over the fitted range; the same optimum was recovered from several different starting points inside the bounds.

2.3 Transient SPM simulation

With the fitted parameters and the effective diffusivity of Eq. (11), the transient PDE was solved with $D = 7 \times 10^{-15} \text{ m}^2/\text{s}$, $R_p = 10 \text{ }\mu\text{m}$, $c_{\text{max}} = 48\,000 \text{ mol/m}^3$, $\rho_{\text{NMC}} = 2.11 \text{ g/cm}^3$, and $i_{\text{app}} = -1.0 \text{ mA/cm}^2$.

Table 2: Fitted NMC111 interaction parameters.

Parameter	Fitted value
V_0	3.824 V
Ω_2^c/F	-0.000 V
Ω_3^c/F	2.000 V
Ω_4^c/F	-2.515 V
Ω_5^c/F	10.000 V
Ω_6^c/F	11.000 V
Ω_7^c/F	-4.000 V
RMSE	22 mV

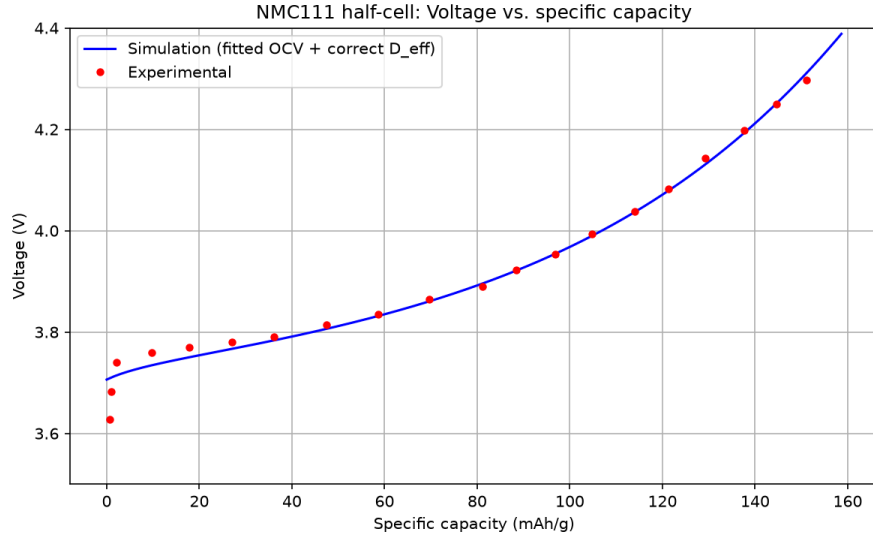


Figure 2: Simulated (fitted-parameter SPM) and experimental (Table 2) voltage vs. specific capacity for the NMC111 half-cell.

3 Full-Cell Voltage vs. Specific Capacity

3.1 Combining the half-cells

The full cell is obtained by running the anode model of Section 1 ($i_{\text{app}} = +1.0 \text{ mA/cm}^2$) and the cathode model of Section 2 ($i_{\text{app}} = -1.0 \text{ mA/cm}^2$) *concurrently over the same physical time*, consistent with the stated full-cell current $i_{\text{app}} = 1.0 \text{ mA/cm}^2$ (positive from cathode to anode). Because the same current density is applied to both electrodes for the same elapsed time, charge is conserved between them automatically – no independent electrode-mass or loading ratio is required beyond the single-particle parameters already given in Problems 1 and 2. The full-cell voltage is then

$$V_{\text{full}}(t) = V_{\text{cathode}}(t) - V_{\text{anode}}(t), \quad (15)$$

reported against the cathode's own specific-capacity trace $Q_{\text{sp}}(t)$ from Eq. (14). The curve is physically meaningful up to the point at which either electrode is exhausted; here the anode (the smaller-capacity electrode, given its particle size and site concentration) is the limiting electrode, and the trace is shown up to $\theta_{\text{anode}} \rightarrow 1$.

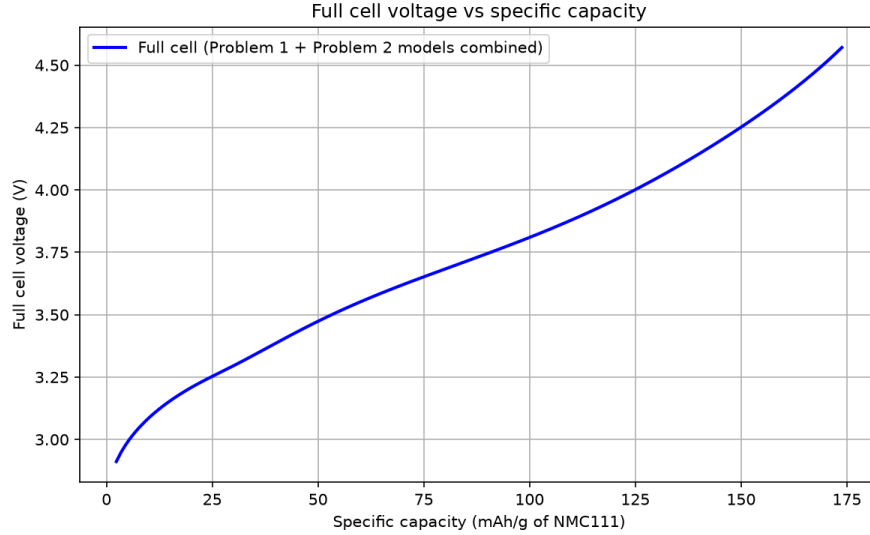


Figure 3: Full-cell voltage vs. specific capacity (NMC111 basis), obtained by combining the anode and cathode SPMs of Problems 1 and 2 at $i_{\text{app}} = 1.0 \text{ mA/cm}^2$.

The full-cell curve (Fig. 3) rises smoothly from about 2.9 V at low capacity to over 4.5 V as the anode approaches saturation, with the characteristic steep upturn near 150 mAh/g to 175 mAh/g that reflects the anode voltage collapsing toward zero as $\theta_{\text{anode}} \rightarrow 1$ – the expected signature of the capacity-limiting electrode in a full cell.

4 Cyclic Full-Cell Voltage with Finite Exchange Current Density

4.1 Butler–Volmer overpotential

For symmetry factor $\beta = 1/2$, the Butler–Volmer relation $i = i_{\text{ex}}[e^{(1-\beta)f\eta} - e^{-\beta f\eta}]$ inverts to

$$\eta = \frac{2RT}{F} \operatorname{asinh}\left(\frac{i_{\text{app}}}{2i_{\text{ex}}}\right). \quad (16)$$

Both electrode/electrolyte interfaces are assigned the same i_{ex} . On *discharge* (Li^+ moving cathode $\rightarrow$ anode reversed relative to Problem 3, i.e. the cell current sourced externally), each electrode reaction is displaced below its equilibrium potential in the direction that opposes the current, so the cell voltage is reduced relative to open circuit by twice the single-electrode overpotential; on *charge* the current reverses and the same overpotential instead *adds* to the open-circuit voltage:

$$V_{\text{discharge}}(Q) = V_{\text{OCV}}(Q) - 2\eta, \quad V_{\text{charge}}(Q) = V_{\text{OCV}}(Q) + 2\eta, \quad (17)$$

where $V_{\text{OCV}}(Q)$ is the equilibrium full-cell curve of Fig. 3 (Section 3, $\eta = 0$). One full cycle is constructed by evaluating Eq. (17) along the discharge leg from $Q = 0$ up to the capacity limit, then back along the reversed state-of-charge path for the charge leg, plotted against cumulative throughput capacity.

4.2 Results

Figure 4 shows the resulting cycle for $i_{\text{ex}} \in \{1, 10, 10^2, 10^3, 10^4\}$ mA/cm² at $|i_{\text{app}}| = 1.0$ mA/cm². As Eq. (16) predicts, $\eta \rightarrow 0$ once $i_{\text{ex}} \gg i_{\text{app}}$, so the $i_{\text{ex}} \geq 10$ mA/cm² curves collapse onto the equilibrium curve and are indistinguishable from one another; only $i_{\text{ex}} = 1$ mA/cm² (comparable in magnitude to i_{app}) shows a visibly open hysteresis loop, with the charge leg running above and the discharge leg below the equilibrium curve by $2\eta \approx 50$ mV near mid-capacity.

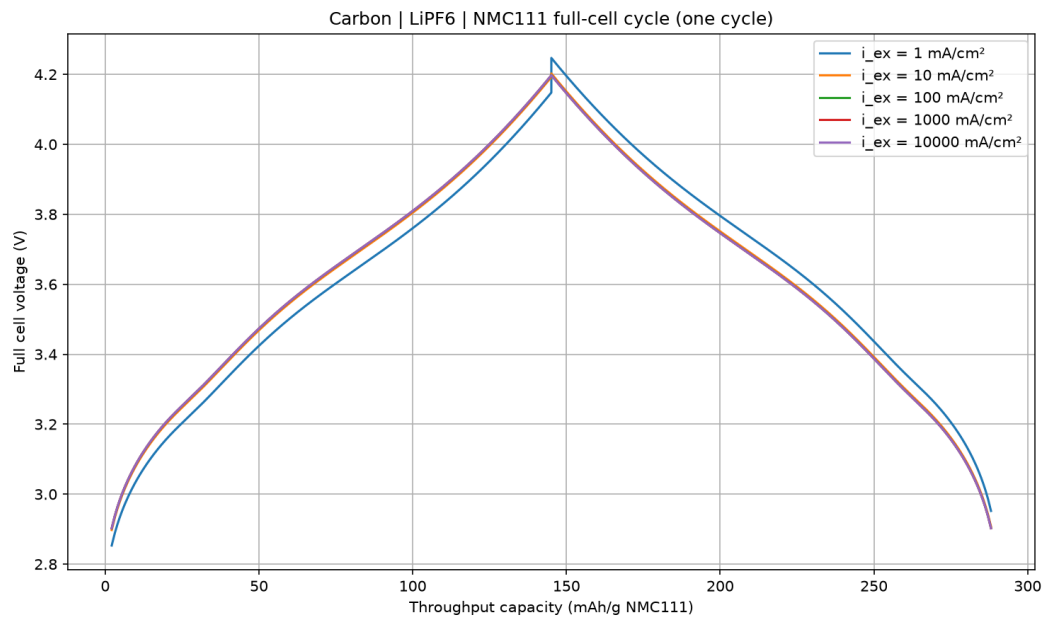


Figure 4: Cyclic full-cell voltage vs. throughput capacity for five exchange current densities, $|i_{app}| = 1.0 \text{ mA/cm}^2$.

Summary

A single-particle model was developed for the Carbon | LiPF₆ | NMC111 cell. The carbon anode half-cell (Section 1) reproduces the Verbrugge–Koch interaction-model voltage–occupancy curve directly from published parameters. The NMC111 half-cell (Section 2) uses an analogous thermodynamic-factor-corrected diffusivity, with its seven open interaction parameters obtained by constrained least-squares fitting to the reported voltage–capacity data (RMSE 22 mV). Combining the two half-cell SPMs under a shared applied current (Section 3) yields a full-cell voltage–capacity curve that correctly reflects the anode as the capacity-limiting electrode. Finally, incorporating Butler–Volmer kinetics at finite exchange current density (Section 4) reproduces the expected narrowing of the charge/discharge hysteresis loop as i_{ex} increases, collapsing to the equilibrium curve once the exchange current density greatly exceeds the applied current density.